

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0278

Hannibal Port Power Project 138 kV
October 30, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0278. The Interconnection Customer proposes a 160.4 MW natural gas combined cycle facility interconnecting at the Hannibal Port Power Project 138 kV substation in the West Virginia area. The request was received on October 20, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 201 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$146,390,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Rec Harmony-Dillard-Orange 345 KV is overloaded to 147.9% of its 1,031 MVA emergency rating for the loss of the Rec Harmony-Oreg 3 Inc Substation 345 KV L. The project contributes 29.1% DFAX (46.6 MW) to this overload.

Grana 1907-1909-Huston 138 KV: 129.8% loading (225 MVA) following the outage of the Dillard-Orange-Rec Harmony 345 KV; DFAX = 12.0%.

For the loss of Rec Harmony-Dillard-Orange 345 KV, the Grana 1907-1909-Pana N 138 KV loads to 126.8% of its 235 MVA rating. Project DFAX: 12.0%.

The Rec Harmony-Pana North 138 KV is overloaded to 124.2% of its 246 MVA emergency rating for the loss of the Rec Harmony-Dillard-Orange 345 KV. The project contributes 12.0% DFAX (19.2 MW) to this overload.

The Whitefish-E Sidney 138 KV is overloaded to 112.4% of its 267 MVA emergency rating for the loss of the Dillard-Orange-Fishkill Plains Substation 345 KV Line. The project contributes 15.2% DFAX (24.3 MW) to this overload.

E Sidney-Semiahmoo 138 KV: 112.0% loading (267 MVA) following the outage of the Fishkill Plains-Dillrd-Orange 345 KV; DFAX = 15.2%.

For the loss of Dillard-Orange-Fishkill Plains 345 KV, the Menlo-Panda Substation 138 KV loads to 108.7% of its 229 MVA rating. Project DFAX: 5.5%.

Northern Steel Casting-Solana 138 KV L: 108.6% loading (239 MVA) following the outage of the Fishkill Plains-Dillrd-Orange 345 KV; DFAX = 5.5%.

Solana Substation-Hampshire Paper 138 KV: 107.4% loading (245 MVA) following the outage of the Dillard-Orange-Fishkill Plains 345 KV L; DFAX = 5.5%.

N Adams-Radisson 138 KV: 106.4% loading (193 MVA) following the outage of the Rec Harmony-Dillard-Orange 345 KV L; DFAX = 17.9%.

Waterton-Iron Grove 138 KV: 102.7% loading (174 MVA) following the outage of the Fishkill Plains-Dillard-Orange 345 KV L; DFAX = 6.9%.

The Rec Harmony Substation-Oreg 3 Inc 345 KV is overloaded to 102.3% of its 1,177 MVA emergency rating for the loss of the Rec Hamony-Dillard-Orange 345 KV. The project contributes 24.3% DFAX (38.9 MW) to this overload.

The North Adams-Rec Harmny 138 KV is overloaded to 101.7% of its 176 MVA emergency rating for the loss of the Rec Harmony 345 kV/138 kV XFMR. The project contributes 33.6% DFAX (53.8 MW) to this overload.

For the loss of Hilltop-Piney Creek Project 500 KV, the Allegheny Energy Units 3 4 & 5-Sugar Hill 345 KV loads to 101.5% of its 1,163 MVA rating. Project DFAX: 8.4%.

The Burwell 500/345kV Auto is overloaded to 101.1% of its 692 MVA emergency rating for the loss of the Allegheny Energy Units 3 4 & 5 345-1 kV XFMR. The project contributes 7.7% DFAX (12.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rec Harmony Substation-Dillard-Orange 345 KV L	Rebuild line with higher-capacity conductor	\$52,924,000	100% (\$52,924,000)
Huston-Grana 1907-1909 138 KV	Rebuild line with higher-capacity conductor	\$10,296,000	100% (\$10,296,000)
Grana 1907-1909-Pana North 138 KV	Rebuild line with higher-capacity conductor	\$3,464,000	100% (\$3,464,000)
Rec Harmony-Pana N 138 KV line	Rebuild line with higher-capacity conductor	\$3,772,000	100% (\$3,772,000)
Whitefish-E Sidney 138 KV	Rebuild line with higher-capacity conductor	\$6,590,000	100% (\$6,590,000)
E Sidney-Semiahmoo 138 KV	Rebuild line with higher-capacity conductor	\$8,298,000	100% (\$8,298,000)
Menlo-Panda Substation 138 KV	Reconductor line	\$2,735,000	100% (\$2,735,000)
Solana-Northern Steel Casting 138 KV	Reconductor line	\$2,099,000	100% (\$2,099,000)
Hampshire Paper-Solana 138 KV	Reconductor line	\$2,053,000	100% (\$2,053,000)
N Adams-Radisson 138 KV	Reconductor line	\$2,451,000	100% (\$2,451,000)
Watrton-Iron Grove 138 KV	Reconductor line	\$1,419,000	100% (\$1,419,000)
Rec Harmony-Oreg 3 Inc 345 KV line	Reconductor line	\$47,893,000	100% (\$47,893,000)
North Adams-Rec Harmny 138 KV	Reconductor line	\$1,265,000	100% (\$1,265,000)
Allegheny Energy Units 3 4 & 5-Sugar Hill 345 KV	Reconductor line	\$9,123,000	3% (\$289,000)
Burwell 500/345 kV Transformer	Replace transformer with higher-rated unit	\$24,748,000	3% (\$842,000)
	Total allocated to project		\$146,390,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.